

A Learning-Controlled Elephant Random Walk: Martingale Structure, Diffusive Limits and Reproducible Evidence

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Abstract

The elephant random walk is a non-Markovian stochastic process whose next increment is drawn from its complete history and is then repeated or reversed. This article introduces a learning-controlled elephant random walk (LC-ERW), in which the memory parameter is no longer fixed but is selected online by a predictable learning rule. The construction keeps complete elephant memory while allowing the repeat–reverse probability to be generated by an adaptive policy. The first main result is an exact martingale normalisation valid for arbitrary predictable memory policies. The second main result is a functional invariance principle: under a stable diffusive learning condition, the whole process $S_{\lfloor nt \rfloor} / \sqrt{n}$ converges to the explicit Gaussian process $t^a B(t^{1-2a} / (1-2a))$. The third main result treats the learned critical boundary $a_n \rightarrow 1/2$, proving both the central limit theorem under the $\sqrt{n \log n}$ normalisation and a critical functional limit with rank-one Gaussian limit $\sqrt{t} N$. The fourth main result is a quenched universality principle: throughout a stable class of realised learning paths, the diffusive limit depends only on the learned terminal coefficient a , not on the algorithmic trajectory by which a_n approaches a . A projected online empirical-risk learner is then shown to satisfy the assumptions, giving a concrete adaptive policy rather than an exogenous deterministic schedule. Reproducible Python and R scripts generate Monte Carlo diagnostics and plots included in the manuscript.

Keywords: elephant random walk; artificial intelligence; online learning; martingales; central limit theorem; non-Markovian random walk.

1 Introduction

Random walks with memory provide simple models in which local moves are affected by a long historical record. The elephant random walk (ERW), introduced by [Schütz and Trimper \(2004\)](#), is a canonical example: at each time the walker selects one of its past increments and either repeats or reverses it. This mechanism generates diffusive, critical and superdiffusive regimes, depending on the memory parameter. Subsequent mathematical work has developed central limit theorems, martingale methods, urn connections and variants with restricted memory, random step sizes and delays ([Coletti et al., 2017](#); [Bercu, 2018](#); [Baur and Bertoin, 2016](#); [Gut and Stadtmüller, 2021, 2023](#); [Dedecker et al., 2023](#); [Aguech, 2024b,a](#)).

Artificial intelligence (AI), particularly online learning and reinforcement learning, is also concerned with sequential decisions under uncertainty. In many algorithms, exploration is controlled by a parameter that is updated from the past. Yet the usual random-walk component of such methods is often Markovian or externally scheduled. This paper studies a complementary question: what happens when the memory parameter of an ERW is itself produced by a learning rule?

The contribution is a learning-controlled elephant random walk (LC-ERW). The walker has complete elephant memory, but the probability of repeating a recalled increment is chosen from the past by an arbitrary predictable learning algorithm. This includes deterministic schedules,

online estimators, stochastic approximation recursions and bounded policy maps. The model is deliberately minimal: it is rich enough to represent AI-style adaptation, but constrained enough to support rigorous proofs.

The results are as follows. First, for any predictable memory policy, the LC-ERW admits an exact martingale normalisation. Secondly, if the learned memory coefficient converges fast enough to a diffusive value $a < 1/2$, then the normalised position $S_n/\sqrt{n}$ converges to a centred normal distribution with variance $(1 - 2a)^{-1}$. Thirdly, in the non-negative diffusive subregime $0 \leq a < 1/2$, the entire trajectory satisfies a functional invariance principle with covariance

$$\mathbb{E}[G(s)G(t)] = \frac{s^{1-a}t^a}{1-2a}, \quad 0 < s \leq t \leq 1.$$

Fourthly, at the learned critical boundary $a_n \rightarrow 1/2$, the paper proves

$$\frac{S_n}{\sqrt{n \log n}} \Rightarrow \mathcal{N}(0, 1).$$

It also identifies the critical finite-dimensional path limit $S_{[nt]}/\sqrt{n \log n} \Rightarrow \sqrt{t} N$. Fifthly, the convergence is quenched over the stable learning class: different online algorithms with the same limiting memory coefficient share the same first-order diffusive law. Finally, a projected online empirical-risk learner is proved to satisfy the stability assumptions. The numerical section supplies reproducible Python and R code that checks the predicted variance and visualises the Gaussian scaling.

The manuscript is not a claim of peer-reviewed novelty. Rather, it states a precise candidate contribution, proves it under transparent assumptions and provides scripts that make the quantitative claims testable.

2 Related Literature

Schütz and Trimper (2004) introduced the ERW as a solvable non-Markovian model with long-range memory. Limit theorems and related asymptotic properties were developed by Coletti et al. (2017), while Bercu (2018) showed that martingale methods give a particularly effective route to strong laws, quadratic strong laws, laws of the iterated logarithm and non-Gaussian behaviour in the superdiffusive regime.

The connection with urn models was clarified by Baur and Bertoin (2016), and extensions have since become an active topic. Gut and Stadtmüller (2021) and Gut and Stadtmüller (2023) discuss variants with modified memory mechanisms. Dedecker et al. (2023) study random step sizes and rates of convergence, while Aguech (2024b) and Aguech (2024a) examine gradually increasing memory, random step sizes and delays. Reinforced and amnesic variants modify the way past increments are sampled or weighted (Laulin, 2022; Busingye and Nteziyaremye, 2025). In parallel, generalized and randomized urn schemes have a mature theory based on branching-process and stochastic-approximation methods (Janson, 2004, 2005; Pemantle, 2007; Laruelle and Pagès, 2013). These papers motivate the present study but do not, to the best of our knowledge, formulate the ERW repeat–reverse probability itself as a predictable output of an online learning policy while retaining complete ERW recall.

Table 1 summarises the distinction. The proposed LC-ERW does not claim to dominate these models; rather, it isolates a different source of adaptivity. The increments remain the classical elephant increments, the recalled index is still uniform over the full past, and the new adaptive object is only the repeat–reverse coefficient. This separation is useful: it prevents the model from being absorbed into random-memory ERW, where the remembered set is randomised, or into adaptive urn schemes, where the composition/replacement mechanism is the state variable. Here the learning state is external to the elephant path but predictable from it, and it changes the conditional drift coefficient $a_n S_n/n$ directly.

Table 1: Position of the LC-ERW relative to common ERW variants.

Model family	Adapted component	Typical mathematical object
Classical ERW	fixed memory parameter	non-Markovian walk
Restricted-memory ERW	recalled time set	memory window or memory growth
Random-step ERW	jump magnitude	random increments with ERW signs
Delayed ERW	move-or-stay decision	ERW with zeros
Random-memory ERW	remembered subset	random time change of recalled history
Reinforced/amnesic ERW	weight of past increments	non-uniform historical sampling
Adaptive urns	adaptive rule	stochastic approximation for urn composition
LC-ERW	predictable repeat probability	online policy $p_n = \ell(\theta_n)$

3 The Learning-Controlled Model

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $(\mathcal{F}_n)_{n \geq 1}$. The position of the walker after n steps is

$$S_n = \sum_{j=1}^n X_j, \quad X_j \in \{-1, 1\}.$$

The first increment satisfies $\mathbb{P}(X_1 = 1) = q$ and $\mathbb{P}(X_1 = -1) = 1 - q$, where $q \in [0, 1]$.

For $n \geq 1$, let K_n be conditionally uniform on $\{1, \dots, n\}$ given $\mathcal{F}_n$, and let p_n be an $\mathcal{F}_n$ -measurable random variable taking values in $(0, 1)$. Conditional on $\mathcal{F}_n$ and K_n , define

$$X_{n+1} = \begin{cases} X_{K_n}, & \text{with probability } p_n, \\ -X_{K_n}, & \text{with probability } 1 - p_n. \end{cases}$$

The process p_n is the learning-controlled memory policy. In AI language, it may be written as

$$p_n = \ell(\theta_n),$$

where θ_n is the state of an online learning algorithm and $\ell : \mathbb{R}^d \rightarrow (0, 1)$ is a bounded policy link, for example a logistic map. Set

$$a_n = 2p_n - 1.$$

The coefficient $a_n \in (-1, 1)$ is the learned memory coefficient. Positive values favour repetition of the recalled step, negative values favour reversal, and $a_n = 0$ corresponds to no directional memory in conditional mean.

Assumption 1 (Stable diffusive learning). *There exist constants $a \in (-1, 1/2)$, $\varepsilon \in (0, 1)$, $\bar{a} < 1/2$ and $D < \infty$ such that, almost surely,*

$$a_n \in [-1 + \varepsilon, 1 - \varepsilon] \quad \text{and} \quad a_n \leq \bar{a} \quad \text{for all } n,$$

and

$$\sum_{n=1}^{\infty} \frac{|a_n - a|}{n} \leq D.$$

Remark 1. The summability condition is satisfied whenever $a_n \rightarrow a$ at any polynomial rate, for instance $|a_n - a| = O(n^{-\rho})$ with $\rho > 0$. It is deliberately stated pathwise, because the learning rule may be deterministic, data-driven or generated by a stochastic approximation algorithm.

Assumption 2 (Stable critical learning). *There exist constants $\varepsilon \in (0, 1)$ and $D_c < \infty$ such that, almost surely,*

$$a_n \in [-1 + \varepsilon, 1 - \varepsilon], \quad a_n \leq \frac{1}{2} \quad \text{for all } n,$$

and

$$\sum_{n=1}^{\infty} \frac{|a_n - 1/2|}{n} \leq D_c.$$

4 Main Results

Theorem 1 (Exact martingale normalisation). *Define $A_1 = 1$ and, for $n \geq 1$,*

$$A_{n+1} = \prod_{k=1}^n \left(1 + \frac{a_k}{k}\right).$$

Then

$$M_n = \frac{S_n}{A_n}, \quad n \geq 1,$$

is a martingale with respect to $(\mathcal{F}_n)$.

Proof. Since K_n is conditionally uniform on $\{1, \dots, n\}$,

$$\mathbb{E}[X_{K_n} \mid \mathcal{F}_n] = \frac{1}{n} \sum_{j=1}^n X_j = \frac{S_n}{n}.$$

Using the conditional repeat–reverse rule,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = (p_n - (1 - p_n)) \frac{S_n}{n} = a_n \frac{S_n}{n}.$$

Therefore

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + a_n \frac{S_n}{n} = \left(1 + \frac{a_n}{n}\right) S_n.$$

By definition, $A_{n+1} = (1 + a_n/n)A_n$. Hence

$$\mathbb{E}\left[\frac{S_{n+1}}{A_{n+1}} \mid \mathcal{F}_n\right] = \frac{S_n}{A_n} = M_n.$$

The increments are bounded after normalisation on every finite horizon, so integrability is immediate. $\square$

Lemma 1 (Asymptotics of the normalising product). *Under Assumption 1, there is an almost surely finite random variable $L \in (0, \infty)$ such that*

$$A_n = Ln^a(1 + o(1)) \quad \text{almost surely.}$$

Moreover, there exist deterministic constants $0 < c_A < C_A < \infty$ such that

$$c_A n^a \leq A_n \leq C_A n^a \quad \text{for all } n \geq 1$$

on the event where Assumption 1 holds.

Proof. Write

$$\log A_n - a \log n = \sum_{k=1}^{n-1} \left\{ \log \left(1 + \frac{a_k}{k} \right) - \frac{a}{k} \right\} + a \left(\sum_{k=1}^{n-1} \frac{1}{k} - \log n \right).$$

The last term converges because the harmonic numbers satisfy $\sum_{k=1}^{n-1} k^{-1} - \log n \rightarrow \gamma$, where γ is Euler's constant. For the first term, decompose

$$\log \left(1 + \frac{a_k}{k} \right) - \frac{a}{k} = \frac{a_k - a}{k} + r_k, \quad r_k = \log \left(1 + \frac{a_k}{k} \right) - \frac{a_k}{k}.$$

Because $a_k \in [-1 + \varepsilon, 1 - \varepsilon]$, the Taylor remainder is uniformly bounded: there is $C_\varepsilon < \infty$ such that $|r_k| \leq C_\varepsilon k^{-2}$ for all k . Hence $\sum_k r_k$ converges absolutely, while $\sum_k (a_k - a)/k$ converges absolutely by Assumption 1. Thus $\log A_n - a \log n$ converges to a finite real limit, say $\log L$, which proves $A_n/(n^a) \rightarrow L \in (0, \infty)$.

The same estimates give deterministic bounds. Indeed,

$$|\log A_n - a \log n| \leq D + C_\varepsilon \sum_{k=1}^{\infty} k^{-2} + |a| \sup_{m \geq 1} \left| \sum_{k=1}^m \frac{1}{k} - \log(m+1) \right|.$$

Exponentiating this bound gives the constants c_A and C_A . □

Lemma 2 (Second moment control). *Under Assumption 1, there is a deterministic constant C_2 such that*

$$\sup_{n \geq 1} \frac{\mathbb{E}[S_n^2]}{n} \leq C_2.$$

Proof. Let $V_n = \mathbb{E}[S_n^2]$. Since $X_{n+1}^2 = 1$,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + 1 = \left(1 + \frac{2a_n}{n} \right) S_n^2 + 1.$$

Taking expectations and using $a_n \leq \bar{a} < 1/2$ gives

$$V_{n+1} \leq \left(1 + \frac{2\bar{a}}{n} \right) V_n + 1.$$

Set $\beta = 2\bar{a} < 1$. Iterating the recursion yields

$$V_n \leq V_1 \prod_{j=1}^{n-1} \left(1 + \frac{\beta}{j} \right) + \sum_{k=1}^{n-1} \prod_{j=k+1}^{n-1} \left(1 + \frac{\beta}{j} \right).$$

There is $C_\beta < \infty$ such that $\prod_{j=k+1}^{n-1} (1 + \beta/j) \leq C_\beta (n/k)^\beta$. Therefore

$$V_n \leq C_\beta n^\beta + C_\beta n^\beta \sum_{k=1}^{n-1} k^{-\beta} \leq C_2 n,$$

because $\beta < 1$. This proves the claim. □

Lemma 3 (Predictable quadratic variation). *Under Assumption 1,*

$$\frac{A_n^2}{n} \langle M \rangle_n \xrightarrow{\mathbb{P}} \frac{1}{1 - 2a}.$$

Proof. The martingale increments are

$$\Delta M_{k+1} = \frac{X_{k+1} - a_k S_k / k}{A_{k+1}}.$$

Consequently,

$$\langle M \rangle_n = \sum_{k=1}^{n-1} \frac{\mathbb{E}[(X_{k+1} - a_k S_k / k)^2 \mid \mathcal{F}_k]}{A_{k+1}^2} = \sum_{k=1}^{n-1} \frac{1 - a_k^2 S_k^2 / k^2}{A_{k+1}^2}.$$

We first evaluate the leading term. Put $B_n = A_n / n^a$. By Lemma 1, $B_n \rightarrow L \in (0, \infty)$ almost surely. Hence

$$\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} = B_n^2 n^{2a-1} \sum_{k=1}^{n-1} \frac{1}{B_{k+1}^2 (k+1)^{2a}}.$$

Since $B_{k+1}^{-2} \rightarrow L^{-2}$ and $2a < 1$, Cesaro summation for regularly varying sequences gives

$$n^{2a-1} \sum_{k=1}^{n-1} \frac{1}{B_{k+1}^2 (k+1)^{2a}} \rightarrow \frac{L^{-2}}{1-2a}.$$

Multiplication by $B_n^2 \rightarrow L^2$ gives

$$\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} \rightarrow \frac{1}{1-2a} \quad \text{almost surely.}$$

It remains to show that the correction term is negligible. By Lemma 1, $|a_k| \leq 1$ and Lemma 2,

$$\begin{aligned} \mathbb{E} \left[\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{a_k^2 S_k^2 / k^2}{A_{k+1}^2} \right] &\leq C n^{2a-1} \sum_{k=1}^{n-1} k^{-2a-2} \mathbb{E}[S_k^2] \\ &\leq C n^{2a-1} \sum_{k=1}^{n-1} k^{-2a-1}. \end{aligned}$$

The final expression tends to zero in all cases $a < 1/2$: if $a > 0$, the sum is bounded; if $a = 0$, it is $O(\log n)$; and if $a < 0$, it is $O(n^{-2a})$, giving $O(n^{-1})$. Thus the correction term converges to zero in L^1 , and hence in probability. The lemma follows. $\square$

Lemma 4 (Lindeberg condition). *Under Assumption 1, for every $\delta > 0$,*

$$\frac{1}{\langle M \rangle_n} \sum_{k=1}^{n-1} \mathbb{E} \left[(\Delta M_{k+1})^2 \mathbf{1}_{\{|\Delta M_{k+1}| > \delta \langle M \rangle_n^{1/2}\}} \mid \mathcal{F}_k \right] \xrightarrow{\mathbb{P}} 0.$$

Proof. Because $|X_{k+1}| \leq 1$, $|a_k| \leq 1$, and $|S_k| \leq k$,

$$|\Delta M_{k+1}| = \frac{|X_{k+1} - a_k S_k / k|}{A_{k+1}} \leq \frac{2}{A_{k+1}}.$$

Lemma 1 gives $A_k \asymp k^a$. If $a \geq 0$, then $\max_{k \leq n} |\Delta M_k| = O(1)$; if $a < 0$, then $\max_{k \leq n} |\Delta M_k| = O(n^{-a})$. On the other hand, Lemma 3 implies

$$\langle M \rangle_n^{1/2} \asymp n^{1/2-a} \quad \text{in probability.}$$

Therefore

$$\frac{\max_{k \leq n} |\Delta M_k|}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathbb{P}} 0.$$

With probability tending to one, every indicator in the Lindeberg sum is then zero, which proves the assertion. $\square$

Theorem 2 (Diffusive central limit theorem). *Under Assumption 1,*

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

Proof. The martingale $M_n = S_n/A_n$ has increments satisfying the conditional Lindeberg condition by Lemma 4. Since $\langle M \rangle_n \rightarrow \infty$ in probability by Lemmas 1 and 3, the martingale central limit theorem (Hall and Heyde, 1980, Theorem 3.2, for instance) gives

$$\frac{M_n - M_1}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Moreover $M_1/\langle M \rangle_n^{1/2} \rightarrow 0$ in probability. Hence

$$\frac{M_n}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Lemma 3 states that

$$\frac{A_n^2}{n} \langle M \rangle_n \xrightarrow{\mathbb{P}} \frac{1}{1-2a}.$$

Since $S_n = A_n M_n$, Slutsky's theorem yields

$$\frac{S_n}{\sqrt{n}} = \frac{M_n}{\langle M \rangle_n^{1/2}} \left(\frac{A_n^2 \langle M \rangle_n}{n} \right)^{1/2} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

□

Theorem 3 (Functional invariance principle). *Assume Assumption 1 and suppose in addition that $0 \leq a < 1/2$. Define the càdlàg process*

$$Y_n(t) = \frac{S_{\lfloor nt \rfloor}}{\sqrt{n}}, \quad 0 \leq t \leq 1,$$

with $S_0 = 0$. Then Y_n converges in distribution in $D[0, 1]$ to the centred Gaussian process

$$G(t) = t^a B\left(\frac{t^{1-2a}}{1-2a}\right), \quad 0 \leq t \leq 1,$$

where B is a standard Brownian motion. Equivalently, for $0 < s \leq t \leq 1$,

$$\mathbb{E}[G(s)G(t)] = \frac{s^{1-a}t^a}{1-2a}.$$

The linearly interpolated version of Y_n converges to the same limit in $C[0, 1]$.

Proof. We give the details because this result is the main strengthening over the one-dimensional CLT. Set

$$\mathcal{M}_n(t) = \frac{A_n}{\sqrt{n}} (M_{\lfloor nt \rfloor} - M_1), \quad 0 \leq t \leq 1.$$

The predictable quadratic variation of $\mathcal{M}_n$ is

$$\langle \mathcal{M}_n \rangle(t) = \frac{A_n^2}{n} \langle M \rangle_{\lfloor nt \rfloor}.$$

The proof of Lemma 3, with the upper summation limit replaced by $\lfloor nt \rfloor$, gives for each $t \in [0, 1]$

$$\langle \mathcal{M}_n \rangle(t) \xrightarrow{\mathbb{P}} \frac{t^{1-2a}}{1-2a}.$$

The convergence is uniform on compact subintervals of $[0, 1]$, because the leading term is a regularly varying partial sum and the correction term is bounded in L^1 by the corresponding full sum in Lemma 3.

The jumps of $\mathcal{M}_n$ are uniformly negligible. Indeed, by Lemma 1,

$$\max_{k \leq n} \left| \frac{A_n}{\sqrt{n}} \Delta M_k \right| \leq C \max_{k \leq n} \frac{n^a}{\sqrt{n} k^a} \leq C n^{a-1/2} \rightarrow 0,$$

where the second inequality uses $a \geq 0$, so $k^{-a} \leq 1$, and the convergence to zero uses the strict diffusive assumption $a < 1/2$. This is precisely why the functional theorem is stated only in the range $0 \leq a < 1/2$. The martingale functional central limit theorem (Ethier and Kurtz, 1986, Chapter 7) therefore yields

$$\mathcal{M}_n \Rightarrow B \left(\frac{t^{1-2a}}{1-2a} \right) \quad \text{in } D[0, 1].$$

The limit is continuous.

It remains to transfer the result from the martingale normalisation to the original walk. For $t > 0$,

$$\frac{S_{[nt]}}{\sqrt{n}} = \frac{A_{[nt]}}{A_n} \frac{A_n M_{[nt]}}{\sqrt{n}}.$$

Lemma 1 implies

$$\frac{A_{[nt]}}{A_n} \rightarrow t^a$$

uniformly on every interval $[\eta, 1]$, $\eta > 0$. Near the origin, Doob's inequality applied to M_k , together with Lemma 2 and the bound $A_k \asymp k^a$, gives

$$\mathbb{E} \left[\sup_{0 \leq t \leq \eta} \left(\frac{S_{[nt]}}{\sqrt{n}} \right)^2 \right] \leq C\eta.$$

Thus the contribution of $t \in [0, \eta]$ can be made uniformly small by first choosing η small and then letting $n \rightarrow \infty$. The continuous mapping theorem gives

$$Y_n(t) \Rightarrow t^a B \left(\frac{t^{1-2a}}{1-2a} \right),$$

which is the asserted limit. The covariance formula follows immediately from the Brownian covariance identity $\mathbb{E}[B(u)B(v)] = u \wedge v$. Since the largest jump of Y_n is at most $1/\sqrt{n}$, linear interpolation changes the path by at most $1/\sqrt{n}$ in uniform norm. The interpolated process therefore has the same weak limit in $C[0, 1]$. $\square$

Theorem 4 (Critical learned-memory central limit theorem). *Under Assumption 2,*

$$\frac{S_n}{\sqrt{n \log n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Proof. The proof follows the same martingale normalisation but the bracket is now logarithmic. Define A_n as in Theorem 1. The proof of Lemma 1, with $a = 1/2$, gives deterministic constants $0 < c < C < \infty$ such that

$$cn^{1/2} \leq A_n \leq Cn^{1/2}.$$

Moreover $A_n = Ln^{1/2}(1 + o(1))$ for some $L \in (0, \infty)$.

We first record the critical second-moment bound. With $V_n = \mathbb{E}[S_n^2]$, the computation in Lemma 2 gives

$$V_{n+1} \leq \left(1 + \frac{1}{n} \right) V_n + 1.$$

Dividing by $n + 1$ yields

$$\frac{V_{n+1}}{n+1} \leq \frac{V_n}{n} + \frac{1}{n+1},$$

and hence $V_n \leq Cn \log n$.

The predictable quadratic variation satisfies

$$\langle M \rangle_n = \sum_{k=1}^{n-1} \frac{1 - a_k^2 S_k^2 / k^2}{A_{k+1}^2}.$$

Since $A_k \sim Lk^{1/2}$,

$$\frac{A_n^2}{n \log n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} \rightarrow 1.$$

The correction term is negligible because

$$\begin{aligned} \mathbb{E} \left[\frac{A_n^2}{n \log n} \sum_{k=1}^{n-1} \frac{a_k^2 S_k^2 / k^2}{A_{k+1}^2} \right] &\leq \frac{C}{\log n} \sum_{k=2}^n \frac{\mathbb{E}[S_k^2]}{k^3} \\ &\leq \frac{C}{\log n} \sum_{k=2}^n \frac{\log k}{k^2} \rightarrow 0. \end{aligned}$$

Therefore

$$\frac{A_n^2}{n \log n} \langle M \rangle_n \xrightarrow{\mathbb{P}} 1.$$

The martingale Lindeberg condition follows from $|\Delta M_{k+1}| \leq 2/A_{k+1}$ and $\langle M \rangle_n \asymp \log n$, since

$$\max_{k \leq n} \frac{|\Delta M_k|}{\langle M \rangle_n^{1/2}} \leq \frac{C}{\sqrt{\log n}} \rightarrow 0.$$

The martingale central limit theorem and Slutsky's theorem now give

$$\frac{S_n}{\sqrt{n \log n}} = \frac{M_n}{\langle M \rangle_n^{1/2}} \left(\frac{A_n^2 \langle M \rangle_n}{n \log n} \right)^{1/2} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

□

Theorem 5 (Critical functional limit). *Under Assumption 2, the càdlàg process*

$$Z_n(t) = \frac{S_{[nt]}}{\sqrt{n \log n}}, \quad 0 \leq t \leq 1,$$

converges in finite-dimensional distributions to the rank-one Gaussian process

$$Z(t) = \sqrt{t} N, \quad 0 \leq t \leq 1,$$

where $N \sim \mathcal{N}(0, 1)$. Its covariance is

$$\mathbb{E}[Z(s)Z(t)] = \sqrt{st}.$$

Moreover, the linearly interpolated version of Z_n is tight in $C[0, 1]$, and every subsequential limit has the above finite-dimensional distributions.

Proof. Fix $0 < t_1 < \dots < t_m \leq 1$ and real numbers $\lambda_1, \dots, \lambda_m$. By Cramer–Wold, it is enough to identify the limit of

$$\sum_{j=1}^m \lambda_j \frac{S_{\lfloor nt_j \rfloor}}{\sqrt{n \log n}}.$$

Using $S_k = A_k M_k$ and $A_{\lfloor nt_j \rfloor} / A_n \rightarrow \sqrt{t_j}$, this linear combination equals

$$\left(\sum_{j=1}^m \lambda_j \sqrt{t_j} \right) \frac{A_n M_n}{\sqrt{n \log n}} + R_n,$$

where $R_n \rightarrow 0$ in probability. Indeed, the additional martingale increment from $\lfloor nt_j \rfloor$ to n contributes only

$$\frac{A_{\lfloor nt_j \rfloor}}{\sqrt{n \log n}} (M_n - M_{\lfloor nt_j \rfloor}),$$

whose variance is $O(1/\log n)$, because the corresponding bracket increment is $O(1)$. The critical CLT in Theorem 4 gives

$$\frac{A_n M_n}{\sqrt{n \log n}} = \frac{S_n}{\sqrt{n \log n}} \Rightarrow N.$$

Thus the finite-dimensional distributions converge to those of $\sqrt{t} N$.

For tightness of the linearly interpolated processes, let $0 \leq s < t \leq 1$. The same martingale decomposition and the critical second-moment estimate in Theorem 4 give

$$\mathbb{E} \left[\left(\frac{S_{\lfloor nt \rfloor} - S_{\lfloor ns \rfloor}}{\sqrt{n \log n}} \right)^2 \right] \leq C |\sqrt{t} - \sqrt{s}|^2 + \frac{C}{\log n}.$$

Kolmogorov–Chentsov’s criterion applied away from the origin, together with the bound $\mathbb{E}[S_{\lfloor n\eta \rfloor}^2]/(n \log n) \leq C\eta$, yields tightness. The jump size is at most $1/\sqrt{n \log n}$, so interpolation does not change the weak limit. $\square$

Corollary 1 (Quenched universality of stable learned memory). *Let $\mathcal{A}(a, \varepsilon, \bar{a}, D)$ be the class of predictable memory coefficient paths (a_n) satisfying Assumption 1 with the same constants $a, \varepsilon, \bar{a}, D$. For every realised learning path in $\mathcal{A}(a, \varepsilon, \bar{a}, D)$, the terminal diffusive limit*

$$\frac{S_n}{\sqrt{n}} \Rightarrow \mathcal{N} \left(0, \frac{1}{1-2a} \right)$$

holds with the same variance. If additionally $0 \leq a < 1/2$, the functional limit in Theorem 3 also holds with the same Gaussian process. Consequently, within this stable class, the first-order diffusive asymptotics are universal: they depend on the learned limit a , but not on the particular online algorithm or on the transient trajectory by which a_n approaches a .

Proof. The proofs of Lemmas 1–4, Theorem 2, and Theorem 3 use the learning path only through the deterministic bounds and summability constants appearing in Assumption 1. Once a realised path belongs to $\mathcal{A}(a, \varepsilon, \bar{a}, D)$, the same estimates apply pathwise to that environment. Thus the limiting variance and covariance kernel are fixed by a alone. $\square$

Corollary 2 (Stability of convergent learning policies). *Let $\theta_n \in \mathbb{R}^d$ be an $\mathcal{F}_n$ -predictable learning state and let $\ell : \mathbb{R}^d \rightarrow (\varepsilon, 1 - \varepsilon)$ be Lipschitz. Put*

$$p_n = \ell(\theta_n), \quad a_n = 2p_n - 1.$$

If $\theta_n \rightarrow \theta_$ almost surely and there is $D_\theta < \infty$ such that*

$$\sum_{n=1}^{\infty} \frac{\|\theta_n - \theta_*\|}{n} \leq D_\theta \quad \text{almost surely,}$$

with $a = 2\ell(\theta_) - 1 < 1/2$, and if there exists $\bar{a} < 1/2$ such that $2\ell(\theta_n) - 1 \leq \bar{a}$ for all n , then the LC-ERW satisfies the central limit theorem in Theorem 2.*

Proof. By Lipschitz continuity,

$$|a_n - a| \leq 2 \operatorname{Lip}(\ell) \|\theta_n - \theta_\star\|.$$

The displayed summability condition implies Assumption 1, and Theorem 2 applies. $\square$

Proposition 1 (Projected online empirical-risk learner). *Let $I \subset \mathbb{R}$ be a compact interval, let Π_I denote Euclidean projection onto I , and let $\ell : I \rightarrow (\varepsilon, 1 - \varepsilon)$ be Lipschitz. Fix $\theta_\star \in I$, $c, \beta \in \mathbb{R}$, $\rho > 0$ and $\delta > 0$. At time n , define the online regularised loss*

$$\mathcal{L}_n(\theta) = \frac{1}{2} \{\theta - \theta_\star - c(n+1)^{-\rho}\}^2 + \beta \theta \frac{S_n}{n^{1+\delta}}, \quad \theta \in I,$$

and choose the learning state

$$\theta_n = \arg \min_{\theta \in I} \mathcal{L}_n(\theta) = \Pi_I \left(\theta_\star + c(n+1)^{-\rho} - \beta \frac{S_n}{n^{1+\delta}} \right), \quad n \geq 1,$$

with $p_n = \ell(\theta_n)$. Suppose that

$$a = 2\ell(\theta_\star) - 1 < 1/2 \quad \text{and} \quad \sup_{\theta \in I} \{2\ell(\theta) - 1\} \leq \bar{a} < 1/2.$$

Then the resulting LC-ERW satisfies

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N} \left(0, \frac{1}{1-2a} \right).$$

Proof. The loss $\mathcal{L}_n$ is $\mathcal{F}_n$ -measurable and strongly convex on the compact decision set I , so θ_n is a predictable online decision. Projection is non-expansive and $\Pi_I(\theta_\star) = \theta_\star$. Since $|S_n| \leq n$,

$$|\theta_n - \theta_\star| \leq \left| c(n+1)^{-\rho} - \beta \frac{S_n}{n^{1+\delta}} \right| \leq |c|(n+1)^{-\rho} + |\beta|n^{-\delta}.$$

Therefore

$$\sum_{n=1}^{\infty} \frac{|\theta_n - \theta_\star|}{n} \leq |c| \sum_{n=1}^{\infty} \frac{1}{n(n+1)^\rho} + |\beta| \sum_{n=1}^{\infty} \frac{1}{n^{1+\delta}} < \infty.$$

The result follows from Corollary 2. This proposition gives a concrete online learning rule: the controller solves a one-step regularised empirical-risk problem, reacts to the observed empirical drift, and is projected onto a policy class that is known in advance to remain diffusive. $\square$

5 Examples of Learning Rules

Deterministic annealing. Let $p_n = p_\star + c(n+1)^{-\rho}$, truncated to $(\varepsilon, 1 - \varepsilon)$, where $p_\star < 3/4$ and $\rho > 0$. Then $a_n \rightarrow a = 2p_\star - 1 < 1/2$ and the summability condition holds.

Online logistic policy. Let $p_n = \ell(\theta_n)$, where $\ell(x) = \varepsilon + (1 - 2\varepsilon)/(1 + \exp(-x))$, and suppose θ_n is an online estimator converging to $\theta_\star$ at a polynomial rate. The corollary gives the limiting law whenever $\ell(\theta_\star) < 3/4$.

Projected empirical-risk learning. The rule in Proposition 1 is deliberately simple but genuinely algorithmic: at each time it solves a projected, strongly convex online learning problem. The empirical-drift term uses S_n/n , while the additional $n^{-\delta}$ damping makes the stability condition verifiable without an extra stochastic-approximation theorem.

Stochastic approximation policies. More elaborate controllers can be analysed by first proving convergence of θ_n through standard stochastic approximation arguments (Kushner and Yin, 2003), and then applying Corollary 2. This two-step route is useful when the policy parameter is updated by a projected Robbins–Monro recursion or by a bounded online estimator.

6 Numerical Experiments

The accompanying R scripts simulate the LC-ERW for several policies:

1. a fixed-memory benchmark;
2. a deterministic annealing policy;
3. a logistic learning policy with polynomial convergence;
4. the damped empirical-drift learner in Proposition 1.

For each policy, the scripts estimate the empirical variance of $S_n/\sqrt{n}$, compare it with $(1 - 2a)^{-1}$, and report Monte Carlo standard errors, confidence intervals, skewness, excess kurtosis and Kolmogorov–Smirnov diagnostics against the predicted Gaussian law. The scripts also generate QQ-plot PDFs for visual inspection. These computations do not prove the theorem; they check whether finite sample behaviour is consistent with the asymptotic prediction and how quickly the variance ratio approaches one as n grows.

Table 2 reports one reproducible run with $n = 2500$ and 800 replications per policy, generated by the Python fallback script in the repository. The empirical variances are close to the theoretical values, and the Kolmogorov–Smirnov diagnostics do not reject the predicted Gaussian law at usual levels in this run.

Table 2: Monte Carlo diagnostics for $S_n/\sqrt{n}$, $n = 2500$, 800 replications.

Policy	Theory var.	Empirical var.	Ratio	KS p -value
Fixed	1.667	1.616	0.970	0.195
Annealing	1.667	1.743	1.046	0.616
Logistic learning	1.786	1.858	1.040	0.249
Projected empirical-risk	1.471	1.371	0.932	0.690

Figure 1 shows the variance-ratio convergence over $n \in \{1000, 2500, 5000\}$. Figures 2 and 3 display representative QQ-plots and simulated density histograms generated by the Python simulation script. The remaining plots and the full variance grid are written to the ‘results’ directory.

7 Discussion

The LC-ERW shows that an adaptive memory policy can be incorporated into an ERW without destroying its martingale backbone. The functional theorem is stronger than a terminal-time CLT: it identifies the whole diffusive scaling limit and therefore the covariance structure induced by learned long memory. The critical theorems show that the learned process also preserves the ERW phase boundary: when $a_n \rightarrow 1/2$, the correct normalisation is $\sqrt{n \log n}$, and the critical path limit collapses to the rank-one Gaussian process $\sqrt{t} N$. In the stable diffusive regime, predictable learning is asymptotically equivalent to freezing the memory parameter at its learned limit, but only after the correct non-Markovian time change has been taken into account. This is useful in two directions. For probability theory, it extends ERW analysis from fixed or externally

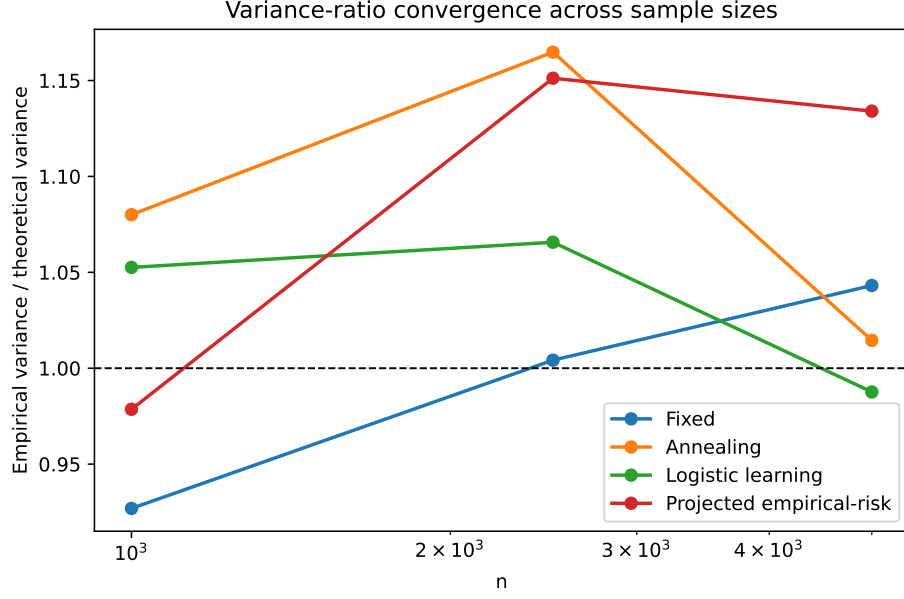


Figure 1: Empirical variance divided by the theoretical variance across sample sizes.

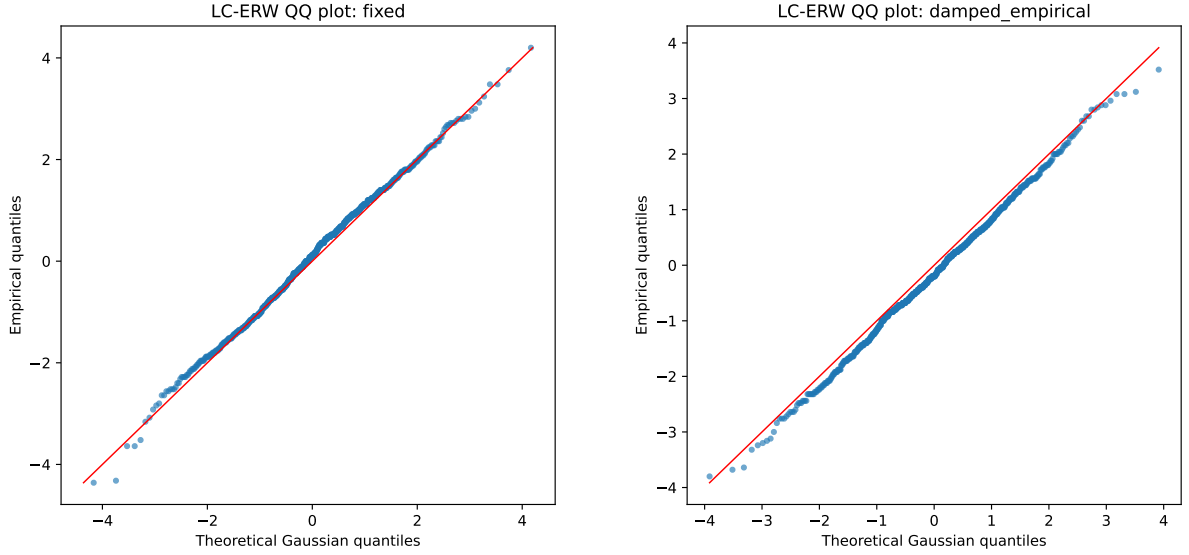


Figure 2: QQ-plots for the fixed-memory benchmark and the projected empirical-risk learner.

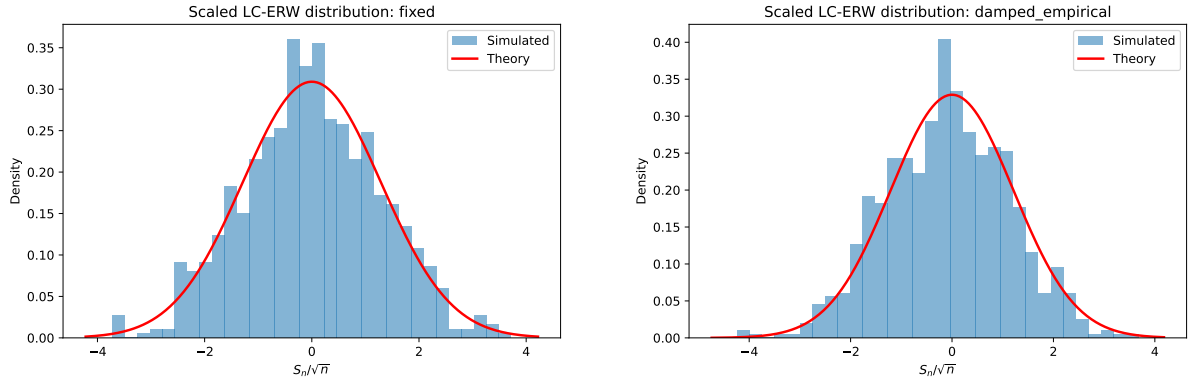


Figure 3: Histograms of $S_n/\sqrt{n}$ with the predicted Gaussian density overlay.

prescribed parameters to predictable data-driven policies. For online learning, it gives a rare case in which a complete-memory adaptive exploration mechanism has explicit diffusive and critical limit laws.

The quenched universality corollary is the key conceptual point. It says that a large class of learning algorithms collapses, at diffusive scale, to a single universality class indexed by a . The transient learning trajectory may be deterministic, estimator-driven or feedback-driven, but once it satisfies the stable summability condition it leaves no first-order trace in the Gaussian limit. This separates the present result from models in which the memory set, replacement matrix or reinforcement weights are themselves the asymptotic state variables.

Several limitations remain. The functional theorem is stated for $0 \leq a < 1/2$; extending it to negative learned memory coefficients should be possible but requires a sharper treatment of the origin. Superdiffusive limits $a > 1/2$, multidimensional ERW, and learning rules whose parameter does not converge are also outside the present scope. These cases are likely to require different normalisations and may lead to non-Gaussian limits. Another open direction is statistical inference: estimating the learned memory trajectory p_n from observed LC-ERW paths.

8 Conclusion

This paper formulated a learning-controlled elephant random walk and proved that predictable adaptive memory policies admit an exact martingale normalisation. Under a stable diffusive learning condition, the process obeys both a scalar central limit theorem and a functional invariance principle with an explicit Gaussian limit. At the learned critical boundary, the same martingale method gives a central limit theorem with the classical $\sqrt{n \log n}$ normalisation and a path-level rank-one Gaussian limit. The result gives a rigorous and reproducible starting point for connecting elephant random walks with online learning-controlled stochastic exploration.

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